Mid Sweden University (Mittuniversitetet)

Omnisound

User-Centred Design and Evaluation
of a Unified Audio Interface

DT168G / IU127G
Human-Computer Interaction / Interaction Design

Project group: Group 1

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Degree programme(s): [TO CONFIRM]
Examiner: [TO CONFIRM]

October 2026

Abstract

TODO: Write an abstract of approximately 250 words after the results and discussion are complete. It should summarise the project's background, purpose, method, results and conclusions.

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Chapter 1

Introduction

1.1 Background

Music, podcasts and audiobooks can form part of the same person's listening activities, while being accessed through different applications. In such a situation, finding audio involves choosing both what to listen to and where to look for it. A listener may save an item in one application, continue a different item elsewhere and later need to remember which application holds each one. These possible organisational difficulties motivate the project.

Finding content is only one part of this problem. Saving an interesting item should also make it possible to find that item again. Similarly, returning to an unfinished podcast or audiobook requires a recognisable route back to the content and its listening position. For example, a student who stops listening to an audiobook during a study break should be able to identify the unfinished book and continue it later.

Combining the three audio types could provide a common place for these activities, but it also raises a design question. A mixed collection may be difficult to navigate if its content types are unclear or if too many items compete for attention. Music tracks, podcast episodes and audiobook chapters also need to remain identifiable within a shared interface. Common navigation and saving controls therefore have to be balanced with clear categories and information about each item.

Existing services already take different approaches to this organisation. Spotify combines music, podcasts and audiobooks in one service, while keeping the content types identifiable within its listening experience [1]. Apple offers separate applications for music, podcasts and audiobooks through Apple Music, Apple Podcasts and Apple Books [2]. These examples show that combining audio content is not a new idea, and neither approach is automatically easier for every user. Omnisound instead explores a specific interface question: whether shared search, a common library and clear content categories can make it easier to find saved material and continue unfinished listening without confusing the different audio types.

Omnisound is the project's response to this question: a web-interface prototype for finding, saving and resuming audio in one organised place. It primarily targets students

who regularly listen to music, podcasts or audiobooks, including listeners who use several content types and those who mainly listen to music. The project investigates how the interface can help these listeners keep track of saved and unfinished content.

The initial high-fidelity prototype (V1) is designed primarily for a desktop browser. It provides a shared search, content-type filters, a library of saved items and a Continue Listening area. Its two main task flows are finding and saving an item, then locating it in the library, and resuming an unfinished podcast or audiobook. These flows give the project a specific interaction focus within the broader audio concept.

1.2 Purpose

The purpose of the project is to investigate how a unified audio interface can support finding, organising and resuming different types of audio content while keeping the experience clear and manageable for the intended users. The motivation is to explore whether a common search and library can reduce the effort of returning to wanted content without making the differences between music, podcasts and audiobooks harder to understand.

This investigation also provides an opportunity to apply user-centred design within the HCI course. The course project requires an iterative process and usability testing in at least one iteration [3, pp. 2–3]. For Omnisound, the purpose is therefore pursued through requirements, alternative designs, prototypes and evaluation. The desired qualities are ease of learning, clear organisation and useful feedback; whether the design supports these qualities must be assessed through the project work.

1.3 Goals

The completed project should provide an identified set of user needs and a prioritised set of interface requirements. The requirements should make clear which listening activities the interface is intended to support and how they relate to the information collected from the target group. This provides a basis for assessing the design beyond its visual appearance.

The design deliverables should include documented alternatives, low-fidelity sketches and interactive high-fidelity prototypes. Alternatives should cover the organisation of audio categories and the presentation of content, such as category tabs compared with a mixed view with filters, and cards compared with lists. The sketches should preserve the early ideas, while the V1 and V2 prototypes should make the selected task flows available for examination. Together, these materials should show how the interface developed and why particular solutions were chosen.

A further deliverable is a documented usability evaluation of the important tasks: finding and saving content, locating saved content and returning to unfinished spoken audio. The evaluation should provide observations of task completion and difficulties, together with an account of how those observations informed design decisions. A record

of design iterations and reasoned conclusions should connect the requirements, prototypes and evaluation. These goals concern identifiable project outputs; they do not assume that all proposed changes will prove effective.

1.4 Delimitations

Omnisound is an interface prototype. The project focuses on navigation, content organisation and the visibility of actions and listening progress. It does not implement music-streaming infrastructure, integration with commercial audio services, payment or subscription processing, licensing systems, backend databases or production account management. The prototype is implemented only as a front end. Limiting the implementation in this way allows the design work to concentrate on the selected interactions within the course project.

Search uses a sample catalogue, and the player controls and displayed progress represent the intended listening experience. They are sufficient to explore how users move through the interface, recognise content and interpret feedback. They do not demonstrate reliable audio delivery, synchronisation between devices or the operation of a real streaming service. Conclusions must therefore concern interaction with the prototype rather than the technical performance of a complete product.

The task scope is also limited. Search and save, retrieval from the library and continuation of unfinished podcasts or audiobooks receive priority. Other possible activities, such as managing a production account or exploring a large commercial catalogue, are outside this evaluation focus. The primary context is desktop use, so the project cannot establish usability across all mobile devices or listening environments. Likewise, conclusions drawn from evaluation with the intended student group should be considered in relation to that group and the tasks tested.

Following the first usability test, a second version of Omnisound was developed to address the problems identified during evaluation. The changes focused on reducing the size of the Home banner, making Continue Listening easier to discover, and improving the clarity of the Save/Saved control. A second usability test was then conducted using V2 and the same three core tasks, allowing the group to compare the two iterations. The testing procedures and findings are presented in the Method and Results chapters. Since Omnisound remains an interface prototype, the conclusions concern the tested interactions rather than the technical performance of a complete streaming service.

1.5 Division of Work

The project was carried out by Obid Alrhman Aljabi, Yacoub Dawli and Sarshar Chavoshi Rad.

Yacoub Dawli was responsible for leading the interviews and moderating the usability tests. Obid Alrhman Aljabi and Sarshar Chavoshi Rad were responsible for documenting

participant responses and observations. During usability testing, Obid focused on participant behaviour and comments, while Sarshar documented task completion times and navigation steps.

The group collaborated on analysing user needs, discussing requirements and evaluating different interface designs. The design process included low-fidelity sketches, two high-fidelity prototypes and usability evaluations. Findings from the first evaluation informed the changes introduced in the second prototype.

The project findings and design decisions were discussed within the group and formed the basis for the report.

Chapter 2

Theory

2.1 User-Centred Design and Iteration

User-centred design begins with understanding the people who will use an interface, what they want to achieve and the circumstances in which they will use it. Rogers, Sharp and Preece explain that understanding people's activities and differences helps designers question assumptions about their needs [4, pp. 16–17]. A target group therefore provides a starting point for research, rather than a reason to assume that all its members have the same habits or expectations.

User needs and goals guide the requirements for an interface. Designers can then explore alternative ways to support those goals and represent the ideas in prototypes. Evaluation with intended users examines whether a proposed solution works as expected. A prototype is a candidate solution that can be tested by observing people using it [5, Sec. "What Is a Testable Prototype?"]. This makes evaluation part of design, rather than a check reserved for the finished product.

Iteration connects evaluation to further design decisions. Findings can lead to revised requirements, a different alternative or changes to an existing interface, followed by another evaluation [5, Sec. "Why Test a Prototype?"]. Simply producing several versions does not establish this connection. A meaningful iteration should explain which problem motivated a change and examine whether the problem remains. This principle is relevant to Omnisound's student audience and its two evaluated high-fidelity prototype iterations.

2.2 Usability and User Experience

ISO 9241-11:2018 relates usability to specified users, goals and contexts of use through effectiveness, efficiency and satisfaction. These definitions are reproduced in ISO/TR 25060:2023 [6, Sec. 3.1]. Usability concerns what people can achieve with a system in a particular situation; it cannot be established from appearance alone.

Effectiveness concerns accurate and complete achievement of a goal, while efficiency concerns the resources used in relation to that outcome [6, Secs. 3.1.7–3.1.8]. In an audio

interface, effectiveness could involve successfully saving an item and finding it again. Efficiency concerns the time and effort required to reach that outcome. Successful completion and a straightforward route are thus different aspects of the same task. Satisfaction concerns how well the experience of use meets the user's needs and expectations [6, Sec. 3.1.9].

Learnability and utility are additional useful HCI concepts. Learnability concerns how easily a person can learn to use an interface, including getting started with unfamiliar controls. Utility concerns whether the available functions support what users need to do [4, p. 21]. For example, a clearly labelled search may be easy to learn, while a way to return to unfinished audio addresses a functional need. These concepts complement the three ISO dimensions; they are not additional components of the ISO definition of usability.

User experience includes people's perceptions and responses before, during and after interaction, including anticipated use [6, Sec. 3.1.15]. Satisfaction and frustration are relevant examples. Task completion alone cannot explain how an interaction feels. Together, these concepts provide a basis for examining searching, saving, retrieving and resuming audio without treating one measure as a complete assessment of the interface.

2.3 Interface Design Principles

Visibility means making the options needed for a task apparent to the user [4, p. 28]. An available function has limited value if people cannot discover it. In an audio interface, recognisable navigation and information about listening progress can help users identify where to go and what can be continued.

Feedback communicates the outcome of an action and the current system state [7, Sec. 1]. After saving an item, a visible confirmation can help the user understand that the action succeeded. Feedback should be timely and clear enough to support the next decision, such as continuing to browse or opening the saved collection.

Consistency means using similar terms, controls and behaviour for similar interactions, both within a product and in relation to familiar conventions [7, Sec. 4]. A shared saving action across different audio types can support predictable interaction. Consistency still allows relevant differences in content, such as episode titles and audiobook chapters.

Affordances concern the actions possible in the relationship between a person and an object. Signifiers are perceivable cues that communicate what can be done [8]. In a graphical interface, a button label, icon or visual treatment can signify an available action. A control's appearance and the action it actually supports should agree. This distinction helps explain why making an action available and making it understandable are separate design concerns.

2.4 Cognition and Mental Models

A mental model is a user's understanding of how a system works. It supports predictions about what an action will do and develops through previous experience [9, Sec. "What Are Mental Models?"]. Users and designers may have different models of the same interface. Familiarity with another audio service may, for example, shape expectations about where saved items belong or whether opening unfinished content will resume it. Clear category names and navigation can help communicate how a shared audio collection is organised.

Attention concerns which information people notice while pursuing a goal. Layout and visual prominence matter because competing information can reduce the visibility of relevant content [7, Sec. 8]. A large introductory area and a route back to unfinished listening may compete for attention on the same screen. This provides a reason to examine the relative prominence of elements, without assuming that size alone determines what every user notices.

Recognition over recall means displaying options and information that users can identify, rather than requiring them to remember details [7, Sec. 6]. Visible content categories, item titles and progress information can support this recognition. For an audio collection, these cues may help someone distinguish saved material from unfinished listening and choose a suitable next action.

2.5 User Research and Evaluation Methods

Qualitative interviews explore people's experiences, needs and expectations. Open questions and follow-up questions allow participants to explain what matters to them in their own terms [10]. Interviews can inform requirements, but they collect accounts of behaviour rather than direct observations of interface use. Combining these accounts with observations can provide a fuller basis for understanding users.

Affinity diagramming organises related observations or ideas into groups that express common themes [11, Sec. "Defining Affinity Diagramming"]. It can help a design team make sense of interview notes and identify connections between needs. The grouping is an interpretation of the material, so observations that differ from a common theme should also remain visible in the analysis.

Personas are research-informed representations of users' goals, needs and expectations. Qualitative personas draw on patterns in research to help a team focus its design decisions [12]. They summarise relevant differences between users rather than reproducing every interview as a separate profile. A persona based only on assumptions should not be treated as evidence about the intended audience, and a qualitative persona does not establish how common a user type is in the wider population.

Low-fidelity prototypes, such as paper sketches, represent early ideas with limited detail. They allow designers to explore layouts and navigation before investing in a detailed interface, and can be changed or discarded readily [5, Sec. "Benefits of Low-Fidelity Prototypes"]. More detailed, interactive prototypes allow evaluation of specific

controls and task flows [5, Sec. “Benefits of High-Fidelity Prototypes”]. The appropriate level of detail depends on the question being investigated; visual polish alone does not establish that a design meets users’ needs.

Think-aloud usability testing asks participants to express their thoughts while completing representative tasks with an interface [13, Sec. “Defining Thinking Aloud Testing”]. Observations of actions, hesitation and comments can reveal difficulties and help explain why a participant chose a particular route. These observations can inform redesign, including changes to the visibility or interpretation of controls.

Qualitative usability tests also have limitations. Speaking aloud creates an unusual situation, and prompts can influence behaviour [13, Sec. “Downsides of Think-Aloud”]. Findings from a small group can identify problems in the tested interactions, but do not automatically provide statistically generalisable conclusions. Task times and navigation steps should therefore be interpreted alongside observations and the test context, rather than treated as universal measures of performance.

2.6 Research Ethics

User research should respect participants’ ability to make an informed and voluntary decision. Vetenskapsrådet’s *God forskningssed 2024* explains that people should understand what participation involves, how information will be handled and how to withdraw consent [14, pp. 62–63]. Information should be understandable to the intended participants, who should have an opportunity to ask questions and decline or stop participation without pressure.

Privacy requires responsible handling of participant information. Only information needed for the study should be collected, and it should be protected appropriately. Removing names alone may not prevent identification when other details can be combined [14, pp. 66–67]. This matters when selecting background details, quotations and observations for a report.

Accurate reporting is also an ethical responsibility. Observations and measurements should be presented honestly, with interpretations and limitations made clear [14, pp. 11–12]. Results that challenge a design should be retained rather than adjusted to support a preferred conclusion. These principles provide the basis for describing the group’s documented ethical procedures in the Method chapter.

Chapter 3

Method

3.1 Overall Design Process

The project combined early concept development, design alternatives, prototyping and two rounds of usability testing. The concept and paper sketches established the intended listening activities and possible ways to organise the interface. These ideas were developed into the first high-fidelity prototype, V1. Its evaluation provided a basis for a targeted redesign, which produced V2. A second usability test then examined the updated interface using the same core task goals. Figure 3.1 summarises this sequence.

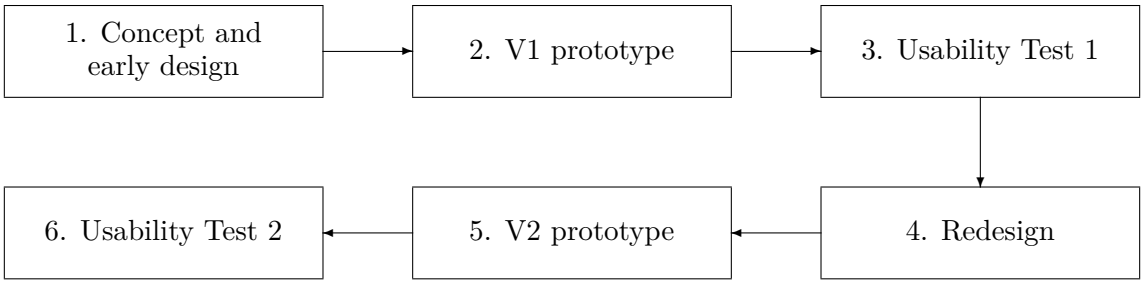


Figure 3.1: Omnisound's design and evaluation process (original project diagram).

This approach was suitable because combining audio types raises questions about how people understand categories, saved content and listening progress. An interactive prototype allowed the group to observe those decisions during use. The user-centred approach described in Section 2.1 therefore informed evaluation of concrete tasks. Iteration connected the observed problems to changes and another evaluation; creating a second version alone would not have provided that connection.

3.2 Initial Design Studio

The early design work was documented through concept material and low-fidelity sketches. The Session 4 concept sheet set out a shared audio interface for students, with finding,

saving and returning to content as its main activities. It also recorded alternative approaches to organisation: separate category tabs compared with a mixed feed and filters, and list views compared with cards. These alternatives framed the question of how to combine content while keeping its different types recognisable.

The original paper sketches explored a navigation area, grouped audio categories and a player layout. Together, they made it possible to examine where navigation, content and playback controls could fit before committing to a detailed interface. They provided concrete material for discussing interface ideas, without requiring each alternative to be fully implemented.

At this stage, the focus was the organisation of activities and information. The concept material connected possible layouts to shared search, a common library and continued listening. Keeping the drawings simple supported consideration of these relationships before visual detail and interactive behaviour became the main focus. The sketches and selected design are presented in the Design chapter.

3.3 Data Collection

To better understand what students might want from a combined audio application, the group prepared an interview plan during Project Session 2. The questions focused on which audio applications people normally used, what they liked about them, what could be improved and what they would expect from an application that brings music, podcasts and audiobooks together. The plan also included showing early interface ideas and asking about first impressions, missing features and how users would expect to complete common tasks.

Yacoub Dawli was assigned to lead the interviews, while Obid Alrhman Aljabi and Sarshar Chavoshi Rad were assigned to take handwritten notes. The questions were mostly open-ended so that participants could explain their habits and preferences in their own words. This approach was useful because the group wanted to understand what people expected from an audio interface before deciding how it should be organised [10].

The available project records include the interview plan but not the completed initial interview responses. For that reason, no specific requirements or findings in this report are attributed to those responses. The later V1 and V2 usability tests also included questions about participants' listening habits, the applications they used and their opinions of the interface. These answers were recorded together with the task observations and helped the group better understand participants' experiences during testing.

3.4 Data Analysis and Requirements

The initial Omnisound concept and the chosen prototype tasks helped the group decide which functions were most important. The interface needed to let users find audio content, save items for later, return to their saved collection and continue listening to something

they had not finished. It was also important that users could easily tell the difference between music, podcasts and audiobooks, even though all three were available in the same application.

Rather than trying to build every feature found in a commercial audio service, the group focused on a few tasks that users could complete from beginning to end. For example, finding an artist was connected to saving that content and checking that it appeared in the Library. Resuming an audiobook required users to find unfinished content and return to the previous listening position. Browsing podcasts was another simple way to examine whether the different audio categories were easy to find. These tasks helped determine which screens, controls and interactions the prototypes needed.

The group also had to decide what could reasonably be included within the scope of a university project. Features such as subscriptions, account management and real audio streaming were left outside the prototype because the main focus was the interface and how people interacted with it. The later usability tests helped identify which parts of the interface were clear and which needed improvement. Although personas and affinity diagrams are discussed as HCI methods in Chapter 2, no completed outputs from those methods are documented in the project material, so they are not presented as steps used in the requirements analysis.

3.5 Prototyping Methods

Low-fidelity sketches and high-fidelity prototypes served different purposes. The paper drawings explored layouts and relationships between navigation, audio categories and playback. They allowed interface ideas to be considered before the group invested in detailed screens. This use of simple drawings is consistent with the value of low-fidelity prototypes for making design changes with limited preparation [5, Sec. “Benefits of Low-Fidelity Prototypes”].

V1 was a desktop-first, browser-based prototype implemented as a front end. Its main screens were Home, Search, Library, content detail and the full player. These screens supported connected task paths rather than only showing isolated layouts. Users could search the sample catalogue, open an item, change its saved state, find it in the Library and open unfinished spoken audio at its displayed progress. This level of interaction allowed the tests to examine labels, navigation choices, controls and feedback.

The catalogue and listening states represented the intended experience. The prototype did not provide a complete streaming service, so opening the player at an existing position was an interface response, not a test of audio delivery. V2 retained this implementation scope and the main task paths. It provided a revised interface for a second evaluation, while V1 remained available as the earlier version. Detailed screen descriptions and visual comparisons belong in the Design chapter.

3.6 Usability Testing

Two usability-testing rounds were conducted with three participants in each round. P1, P2 and P3 tested V1, while P4, P5 and P6 tested V2. All six were students, aged 20–26, who reported regular use of audio applications. Their backgrounds included mainly music listening and combinations of music, podcasts and audiobooks. These characteristics were relevant to the intended student audience. The records describe participants' listening habits but do not specify how they were recruited.

The evaluations used the interactive prototypes in a browser on a laptop, within the project's desktop-use focus. The V1 plan identifies Mittuniversitetet in Sundsvall as the location and specifies that the prototype ran locally. The V2 plan also specifies a laptop and browser. Participants' usual mobile or computer use was recorded as background information; the sessions did not constitute a comparison between devices.

The group divided moderation and observation between its members. Yacoub Dawli led the sessions and presented the tasks. Obid Alrhman Aljabi documented participant behaviour and comments, while Sarshar Chavoshi Rad recorded task times, clicks and navigation observations. This division allowed the moderator to attend to the participant while the observers recorded the interaction.

Each session began with an introduction to the university project and the purpose of the test. After information about participation and consent, the moderator asked which audio applications the participant normally used, which audio types they listened to, how often they listened and whether they normally used mobile devices, computers or both. The three tasks were then presented individually. Participants were asked to explain what they were looking for, what they expected to happen and anything they found unclear while using the interface. This followed the think-aloud approach introduced in Section 2.5 [13, Sec. "Defining Thinking Aloud Testing"].

The instructions described the intended goal without specifying the control to select. For example, the unfinished-audiobook task did not tell participants to use Continue Listening. This left their choice of route available for observation. When prompting was needed, the moderator asked neutral questions about what the participant was thinking or expected. Participants were encouraged to try what made sense to them instead of being immediately directed to a solution.

Table 3.1 summarises the three core tasks and the completion criteria recorded in the test documentation. The first task included checking that the saved item could be found again. The second concerned returning to an existing listening position, while the third examined access to one content category. These tasks covered the main interaction priorities without requiring every prototype function to be tested.

The same core goals and completion criteria were used in the second round so that the group could compare the two versions without replacing the activities being examined. The wording allowed different valid routes to the goal. Finding content through a category, for example, could still meet the first task's criterion. The records could therefore distinguish task completion from the particular route chosen.

Table 3.1: Core task goals and completion criteria used in both testing rounds.

Task	Goal	Completion criterion
1	Find and save Weezer for later.	The item is saved and can be found in the Library.
2	Find an unfinished audiobook and continue listening.	The player opens at the existing progress position.
3	Find where to browse podcasts.	The Podcasts category is reached.

The task records covered completion, elapsed time in seconds, click counts and navigation paths, alongside notes on hesitation, incorrect actions and any need for help. Relevant think-aloud comments accompanied these observations. Completion related to effectiveness, while time, clicks and routes provided information about effort and efficiency. Hesitation and comments helped examine whether controls were discoverable and whether their feedback was understood. These records supported interpretation of the measured values; a time or click count alone did not explain a difficulty.

Post-test questions covered the overall impression, easy or confusing parts, unclear labels, missing functions and suggested changes. The V2 records also included questions about Save and discoverability. The group reviewed these responses alongside the task observations in each round. This connected participants' impressions to what happened during interaction. Individual accounts, measurements and the comparison of results are presented in the Results chapter.

3.7 Iterative Evaluation

After the first evaluation, the group brought together task records, navigation paths and post-test comments to identify issues for redesign. The V1 analysis linked each identified issue to supporting observations, its effect on the tested task and a proposed action. The prioritised problem list distinguished concerns about reaching or discovering task-related content from smaller concerns about visual clarity. It also identified parts of the interface to retain.

The resulting redesign focused on Continue Listening visibility, the height of the Home introductory area and the clarity of the Save/Saved control. These areas connected the recorded concerns to specific changes that could be examined in another test. The basic navigation, categories, Library and core task goals were kept consistent. This limited the scope of the redesign and retained the main interaction context for comparison.

V2 was then evaluated with new participants using the procedure and tasks described in Section 3.6. Using different participants avoided repeating the tasks with people who had already learned the V1 paths. The second-round analysis considered the same completion criteria, measures, routes and participant feedback, including observations related to the targeted changes.

This comparison concerned the tested prototype interactions. With three different participants per round, differences between the records cannot be attributed solely to the redesign or generalised to all students. The purpose was to examine the revised design against the earlier concerns, without assuming that changing the interface established an improvement.

3.8 Research Ethics

The participant introductions explained that the group was evaluating the interface, not the person using it. They described the university project, the observation and written note-taking, and the intended use of the information. Participants were told that taking part was voluntary, that they could stop at any time and that they could raise questions or discomfort. The moderator then asked whether they were comfortable participating, and the records include agreement before the tasks. This approach addressed informed and voluntary participation as discussed in Section 2.6 and in Vetenskapsrådet's guidance [14, pp. 62–63].

The introductions stated that written notes would be anonymous and used for the university project. The retained records identify participants as P1–P6 instead of using personal names. Background information about listening habits and device use provided context for the tests without requiring names in the report.

Accurate reporting also informed the treatment of the material. The testing records distinguish observations and participant comments from the group's interpretations. This distinction is maintained when describing the method and reporting the results, consistent with the responsibility to communicate research honestly [14, pp. 11–12].

Chapter 4

Design

4.1 Initial Design Alternatives

The early concept considered separate audio categories and a mixed feed with filters. Categories give listeners a clear starting point when they already know which audio type they want. A mixed feed supports browsing different types together, but needs clear labels and filters to distinguish them.

Cards and lists offered another choice. Cards leave room for artwork and visual recognition, but fewer items fit on the screen. Lists show titles, creators and other information more compactly. Their suitability depends on whether the user is browsing or looking for a particular item.

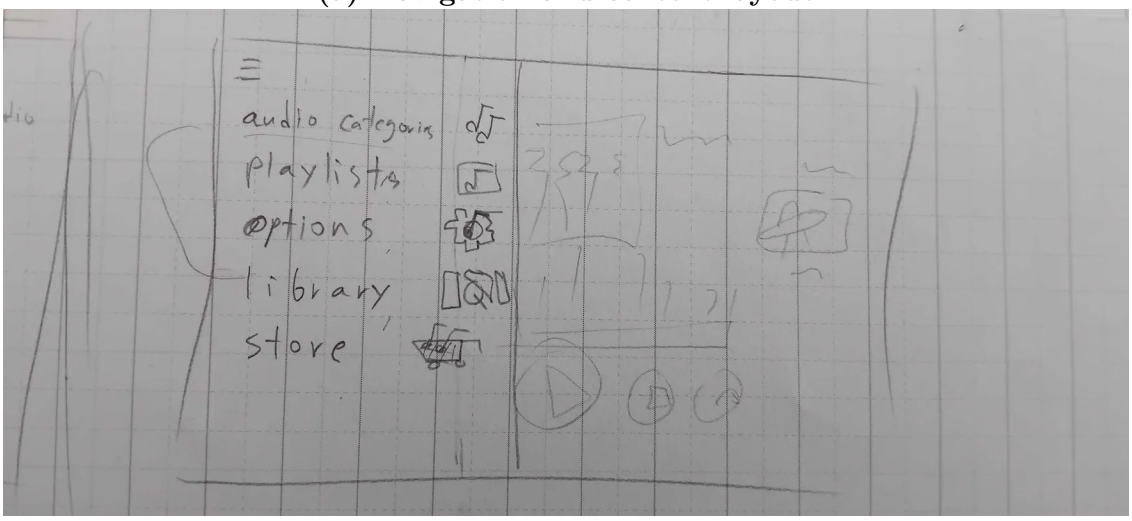
The chosen interface combines these ideas: cards in separate Home sections, a list of search results and one Library with content filters. Each audio category has a browsing route, while saving follows a shared pattern. This gave us a starting design that we could test and improve.

4.2 Lo-Fi Sketches

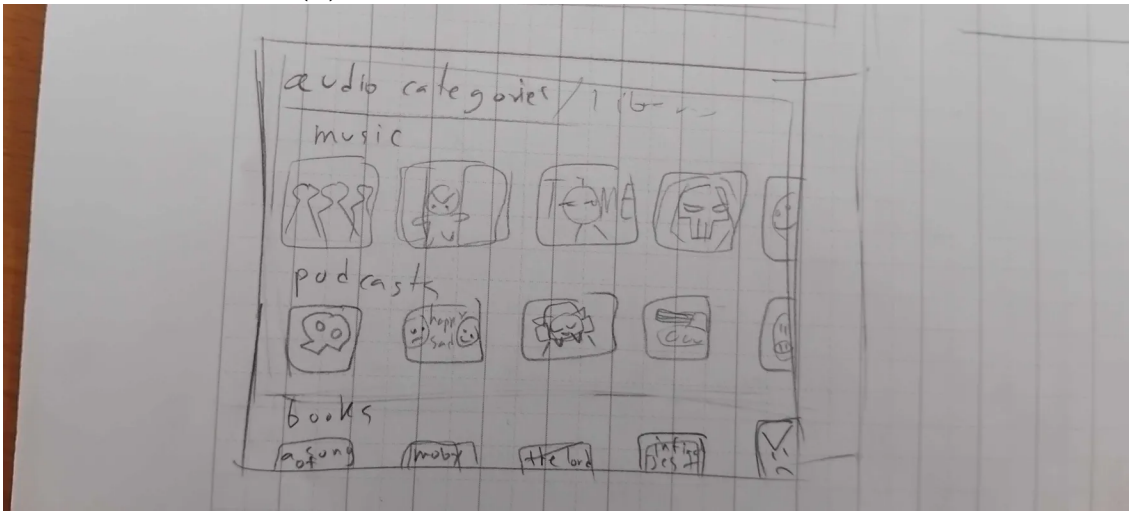
Figure 4.1 presents the group's three original pencil sketches. Sketch (a) places a labelled navigation area on the left, with entries for audio categories, playlists, options, Library and a store. The area on the right contains rough content blocks and playback symbols. Sketch (b) concentrates on browsing: rows labelled music, podcasts and books contain small drawings of covers. Sketch (c) explores a player around Weezer, with central artwork, a large play symbol, neighbouring controls and a horizontal volume control below.

These drawings made the layout ideas visible before the interactive screens were built. They helped the group consider how much space navigation, content and playback would need. Their rough appearance also left room to change the arrangement without first producing detailed visual designs. They were starting points rather than specifications: for example, the store shown in sketch (a) was not included in the tested prototype.

(a) Navigation and content layout



(b) Audio categories and cover cards



(c) Player layout

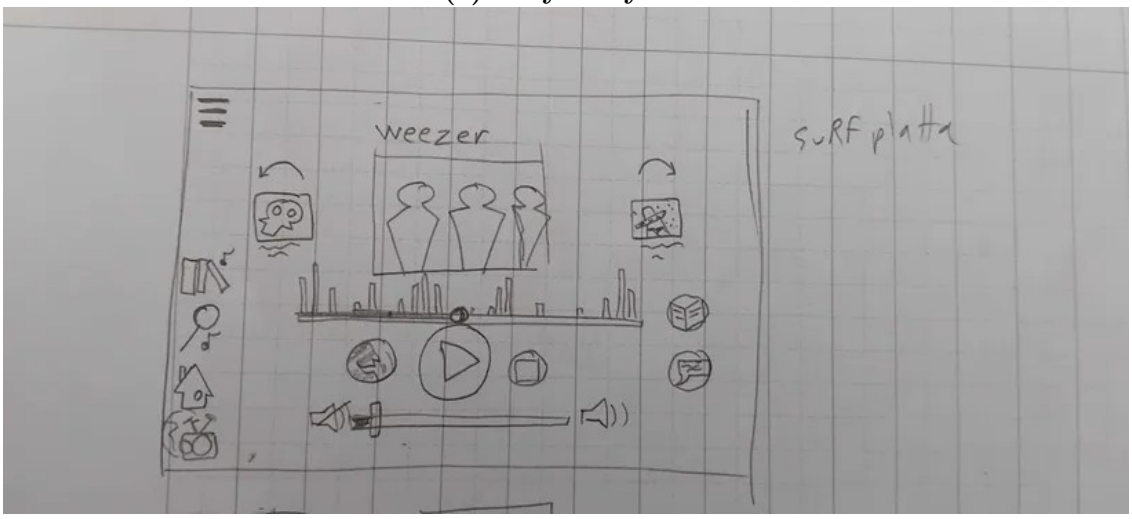


Figure 4.1: The three original project sketches, exploring navigation, category rows and playback. The photographs retain their original content and proportions.

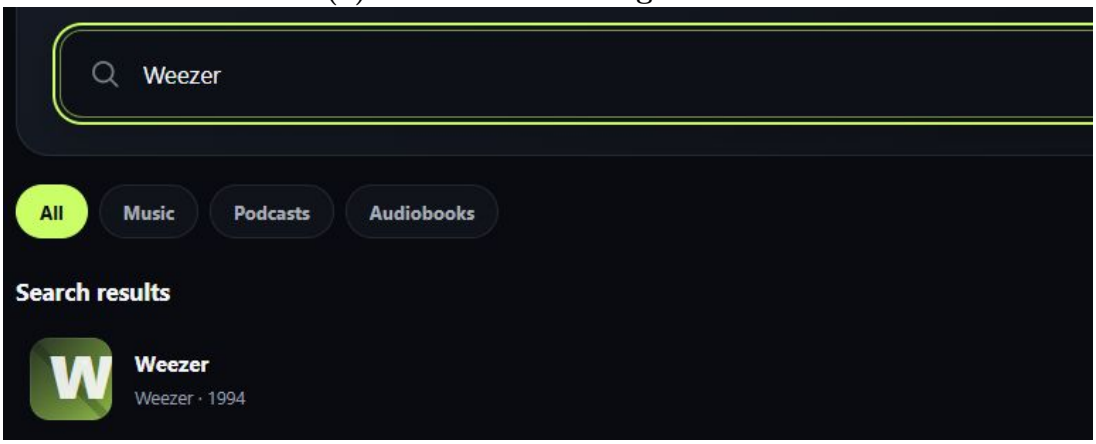
4.3 Omnisound Interface Structure

Omnisound has five main screen types: Home, Search, Library, content detail and the full player. Persistent navigation provides Home, Search and Library, followed by separate Music, Podcasts and Audiobooks routes. These remain available when opening content or playback.

Home includes Continue Listening for unfinished spoken audio. Search covers the sample catalogue, with type filters. Library collects saved items and offers similar filters. Details provide Save and Play or Resume. The full player groups listening controls; a smaller player remains available during browsing after playback opens.

Figure 4.2 shows Weezer in Search and, after saving, in Library. Results use lists for scanning, while Library uses covers and titles for recognition. Shared destinations and controls connect the audio types without requiring different saving and playback patterns.

(a) V1 Search: finding Weezer



(b) V1 Library: Weezer after saving

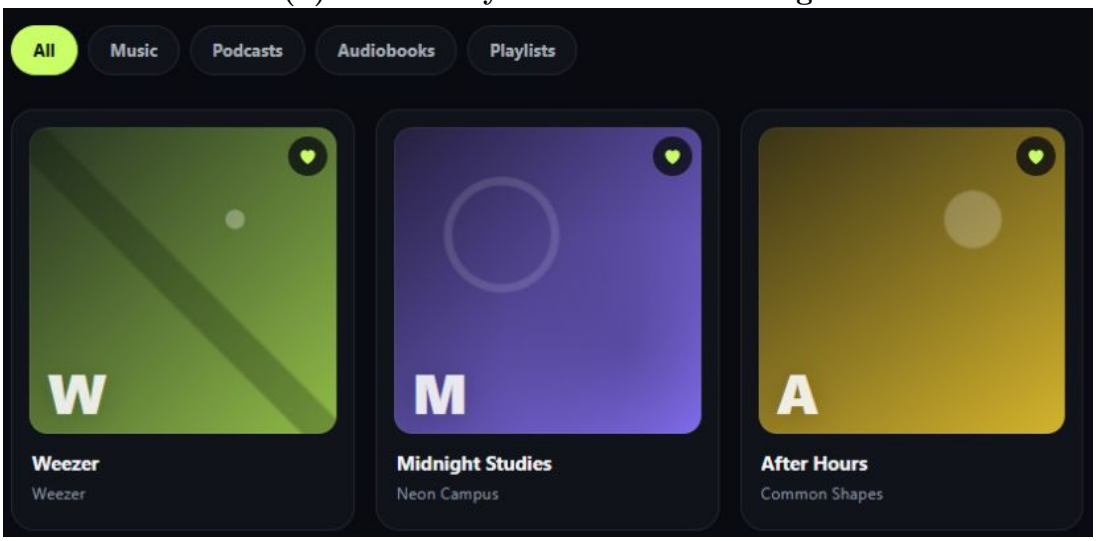


Figure 4.2: V1 Search and Library, cropped to show the query, filters, result and saved collection.

4.4 Hi-Fi Prototype V1

V1 developed the sketches into the first interactive high-fidelity prototype. Instead of viewing isolated drawings, users could follow the main activities from one screen to another. The category rows and artwork remained part of the design, while labelled buttons and visible states made searching, saving and resuming available for testing.

The Home screen in Figure 4.3 starts with a large introductory banner. Below it, a small Continue Listening label sits above the heading "Back in progress". The unfinished items show their audio type, title, creator and listening progress. Separate Music, Podcasts and Audiobooks sections follow further down the page. The sidebar provides a direct route to each category even when its Home section is below the visible area.

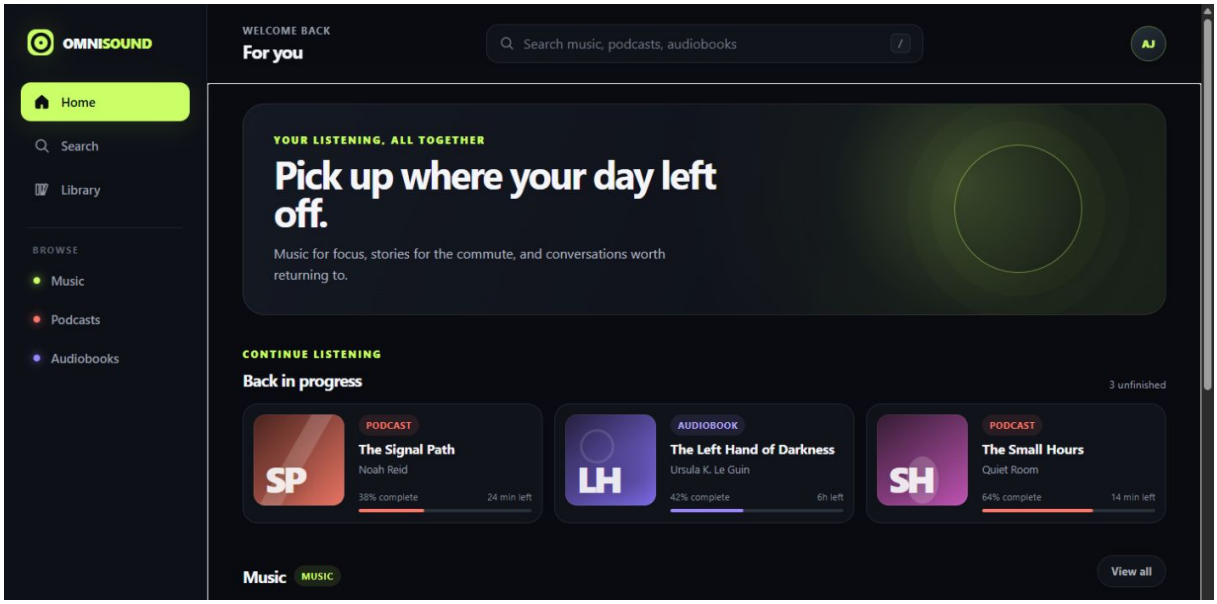
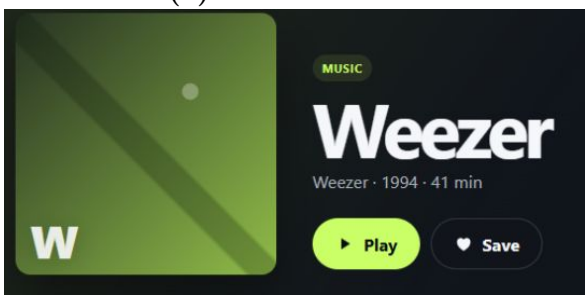


Figure 4.3: The V1 Home screen, showing the persistent navigation, large introductory banner and unfinished content. The capture shows the upper part of Home; further category rows continue below.

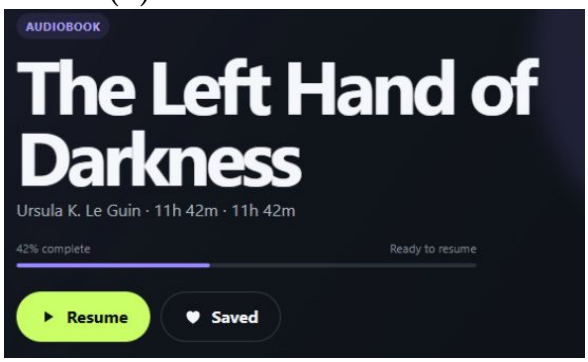
One important workflow begins by searching for Weezer and opening the result. Its detail screen places Save beside Play, as shown in Figure 4.4(a). Selecting Save changes the label to Saved and briefly displays "Weezer saved to your Library". Opening Library then shows the saved item, as in Figure 4.2(b). The connection matters because saving should provide a route back to the content, rather than only changing a button on the current screen.

The other main workflow returns to unfinished audio. A user can open *The Left Hand of Darkness* from Continue Listening and recognise its existing progress on the detail screen. Resume then opens the full player at that displayed position. Figure 4.4(b–c) shows these connected states. The player includes progress, play/pause, volume and spoken-audio speed controls. Its sample content and progress represent interface states; they do not demonstrate a complete streaming service.

(a) Weezer detail



(b) Unfinished audiobook



(c) Full player after Resume



Figure 4.4: V1 saving and resuming workflows: Save beside Play on the Weezer detail page, progress and Resume on the audiobook detail page, and the resulting full-player view. Crops focus on the relevant content and controls.

4.5 Design Decisions and HCI Connections

The principles in Section 2.3 guided concrete parts of the interface. The sidebar labels and categories in Figure 4.3 support visibility. Labels such as Music and Audiobook explain what an item is before it is opened, without relying only on artwork or colour.

Feedback connects actions to outcomes. Saving changes the label and displays a confirmation naming Library. Navigation highlights show the current destination, while play/pause controls reflect the player state. However, clear feedback after saving does not ensure that Save is easy to find beforehand, as the first test showed.

Consistency links the audio types. Their cards open similarly arranged detail pages, with Save beside Play or Resume. The lists below contain tracks, episodes or chapters as appropriate, while the main interaction pattern stays familiar. This connects expectations and behaviour as discussed in Section 2.4.

Recognition also shaped the screens. Titles and covers help users identify Library items, and progress bars identify unfinished audio. These cues reduce the need to remember exact titles or listening positions. Buttons such as Play, Resume and Save use familiar icons and clear labels to show users what they do. Play and Resume state what will happen; the heart with Save gives the icon a written meaning. The redesign addressed cues that needed more emphasis.

4.6 Redesign Decisions

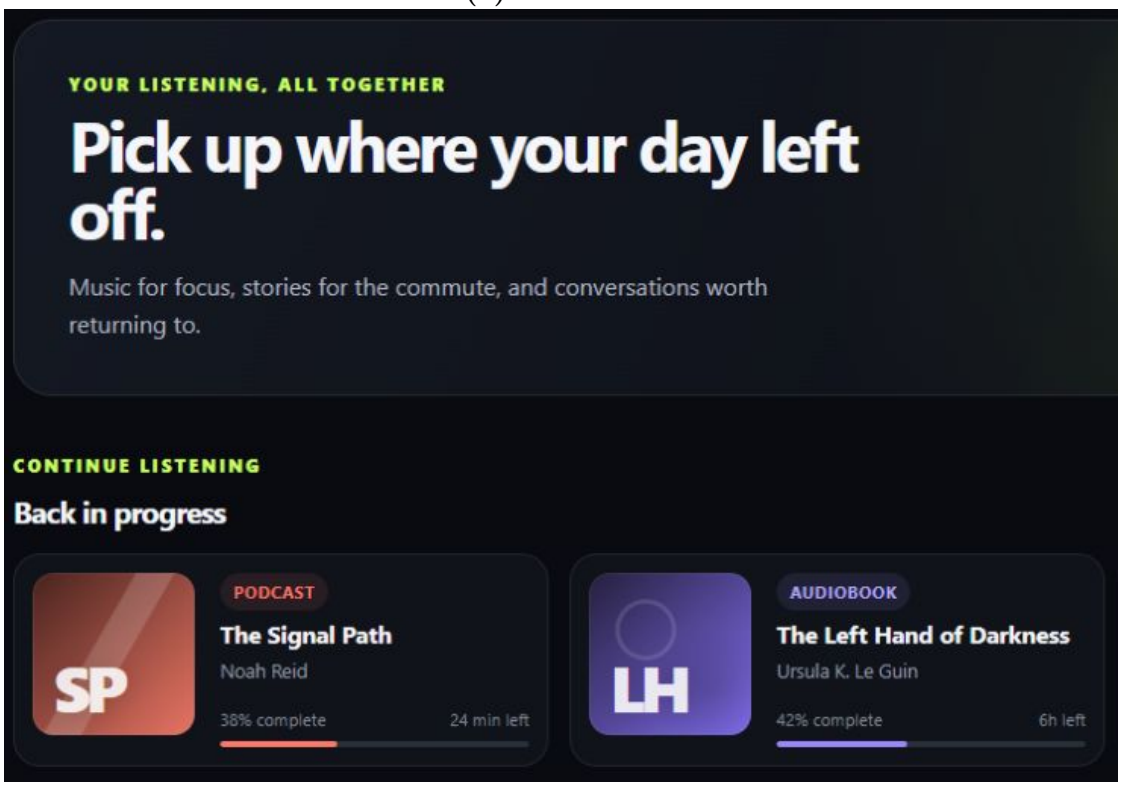
The first usability test showed that some participants looked for unfinished audio through Audiobooks or Library instead of Continue Listening. Those routes worked, but the intended Home shortcut was being overlooked. The group therefore focused the redesign on making that section easier to notice, reducing the space taken by the introduction and clarifying the saving control.

Figure 4.5 shows the Home changes. V2 reduces the banner's height and the space before unfinished content. It also replaces the small Continue Listening label and "Back in progress" heading with a single, larger Continue Listening heading and a lime marker. The unfinished cards and their progress information remain. The intention was to bring attention to content that users could return to, while keeping a short introduction to Omnisound.

The Save control received a smaller change. The first test recorded a brief hesitation in locating it beside Play, so V2 strengthened its border and gave the unsaved and saved states more distinct appearances. The label and confirmation message were retained because they explained the action's result. Figure 4.6 shows the revised states.

The group kept the sidebar, Search, Library and audio categories because the first evaluation supported these routes. Library also provided another way to reach unfinished content. We then tested V2 to see whether these changes made the interface easier to use.

(a) V1 Home



(b) V2 Home

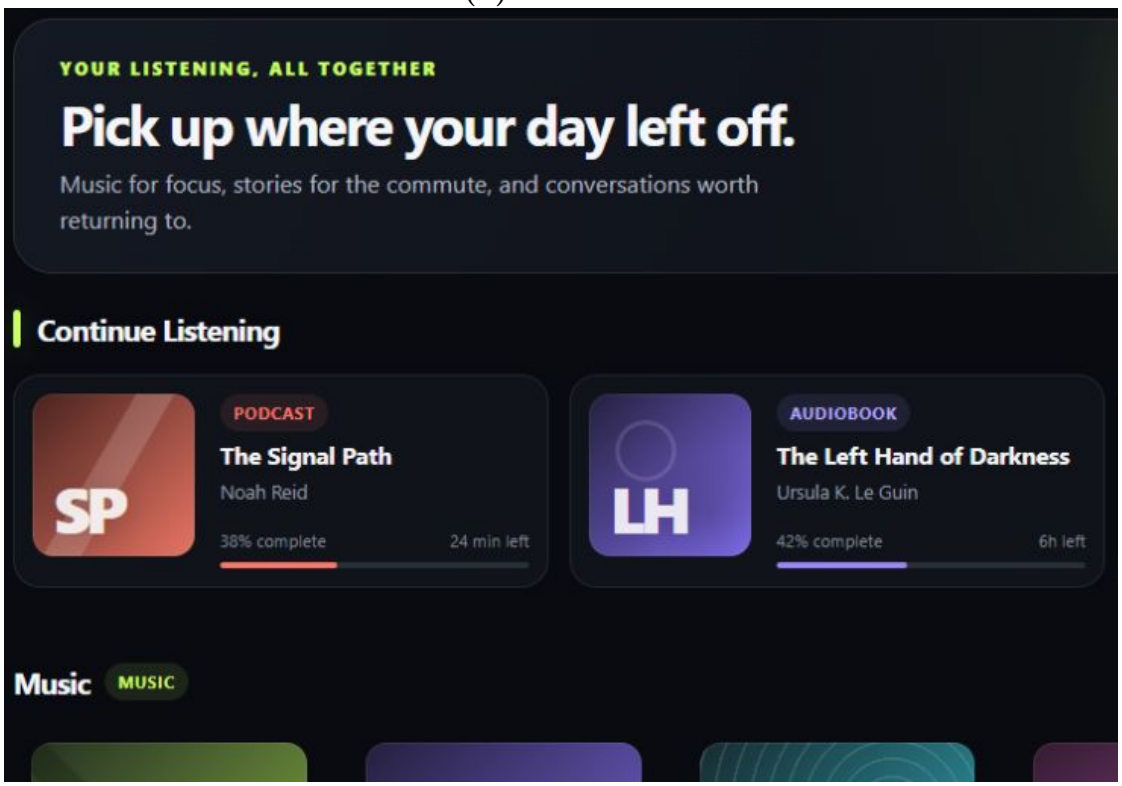


Figure 4.5: Home before and after the redesign. Both prototypes were captured at a 1440×900 viewport and 100% zoom, then cropped to the same region of the banner and first two unfinished cards, at the same scale. The smaller banner and revised heading bring unfinished content higher on V2 Home.

4.7 Hi-Fi Prototype V2

V2 keeps the five main screen types and navigation structure of V1. The Home comparison in Figure 4.5 also shows the final arrangement: the introduction occupies less space, and the more direct Continue Listening heading gives unfinished content greater emphasis. Users can still search, browse categories or open Library through the same destinations.

On the detail page, the unsaved control uses an outlined heart and the label Save. After saving, it shows a filled heart, lime styling and Saved. The comparison in Figure 4.6 captures the same Weezer page before and after selecting Save. The control stays beside Play, so its position remains familiar while its changed appearance communicates the current state. The brief Library confirmation is also retained, giving immediate feedback as well as a state that remains visible on the page.

(a) Before saving **(b) After saving**

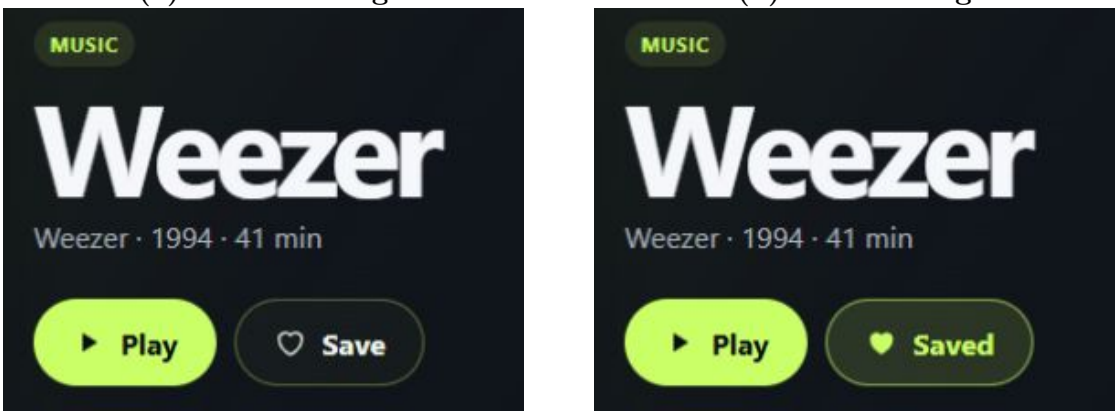


Figure 4.6: The Save and Saved states in V2, captured before and after saving the same item. Matching crops preserve the detail context; only the saving control changes. The temporary confirmation has disappeared in the second capture.

The group evaluated V2 in a second usability test using the same core tasks: finding and saving content, resuming unfinished audio and browsing podcasts. This completed the sequence from V1 evaluation to redesign and another test. Chapter 5 presents the recorded outcomes and comparison between the two rounds.

Chapter 5

Results

TODO: Chapter not yet written.

Chapter 6

Discussion

TODO: Chapter not yet written.

References

[1] Spotify, "Spotify expands audiobook access for premium subscribers in 5 more European countries," Spotify Newsroom, Nov. 2025, published November 18, 2025. Accessed October 13, 2026. [Online]. Available: https://newsroom.spotify.com/2025-11-18/audiobooks-in-premium-sweden-denmark-finland-iceland-monaco/

[2] Apple, "Apps by Apple," Apple Official Website, accessed October 13, 2026. [Online]. Available: https://www.apple.com/apps/

[3] J. Åhlander, "Projekt – användargränssnitt: Människa-datorinteraktion," Mid Sweden University, course project instructions, Oct. 2023, dated October 10, 2023; local file: P3 Projekt.pdf.

[4] Y. Rogers, H. Sharp, and J. Preece, *Interaction Design: Beyond Human-Computer Interaction*, 6th ed. Wiley, 2023, chapter 1 publisher excerpt. Accessed October 13, 2026. [Online]. Available: https://media.wiley.com/product_data/excerpt/9X/11199010/111990109X-20.pdf

[5] K. Pernice, "UX prototypes: Low fidelity vs. high fidelity," Nielsen Norman Group, Dec. 2016, published December 18, 2016. Accessed October 13, 2026. [Online]. Available: https://www.nngroup.com/articles/ux-prototype-hi-lo-fidelity/

[6] International Organization for Standardization, "Systems and software engineering—systems and software quality requirements and evaluation (SQuaRE)—general framework for common industry format (CIF) for usability-related information," International Organization for Standardization, Technical Report ISO/TR 25060:2023, 2023, public preview, Section 3.1. Accessed October 13, 2026. [Online]. Available: https://www.iso.org/obp/ui?_escaped_fragment_=iso:std:iso:tr:25060:ed-1:v1:en

[7] J. Nielsen, "10 usability heuristics for user interface design," Nielsen Norman Group, Apr. 1994, published April 24, 1994; last reviewed January 30, 2024. Accessed October 13, 2026. [Online]. Available: https://www.nngroup.com/articles/ten-usability-heuristics/

[8] D. Norman, "Signifiers, not affordances," JND.org, author's version of an article in ACM Interactions, vol. 15, no. 6, Nov. 2008, posted November 17, 2008. Accessed October 13, 2026. [Online]. Available: https://jnd.org/signifiers-not-affordances/

[9] M. Chan, “Mental models,” Nielsen Norman Group, Jan. 2024, published January 26, 2024; last reviewed September 1, 2026. Accessed October 13, 2026. [Online]. Available: https://www.nngroup.com/articles/mental-models/

[10] M. Rosala and K. Pernice, “User interviews 101,” Nielsen Norman Group, Sep. 2023, published September 17, 2023; last reviewed July 15, 2026. Accessed October 13, 2026. [Online]. Available: https://www.nngroup.com/articles/user-interviews/

[11] R. Krause and K. Pernice, “Affinity diagramming for collaboratively sorting UX findings and design ideas,” Nielsen Norman Group, Apr. 2024, published April 26, 2024; last reviewed August 7, 2026. Accessed October 13, 2026. [Online]. Available: https://www.nngroup.com/articles/affinity-diagram/

[12] P. Laubheimer, “3 persona types: Lightweight, qualitative, and statistical,” Nielsen Norman Group, Jun. 2020, published June 21, 2020. Accessed October 13, 2026. [Online]. Available: https://www.nngroup.com/articles/persona-types/

[13] J. Nielsen, “Thinking aloud: The #1 usability tool,” Nielsen Norman Group, Jan. 2012, published January 15, 2012. Accessed October 13, 2026. [Online]. Available: https://www.nngroup.com/articles/thinking-aloud-the-1-usability-tool/

[14] Vetenskapsrådet, “God forskningssed 2024,” Vetenskapsrådet, Tech. Rep. VR2405, 2024, accessed October 13, 2026. [Online]. Available: https://www.vr.se/analys/rapporter/vara-rapporter/2024-10-02-god-forskningssed-2024.html